

MXCuBE at the Brazilian Synchrotron Light Laboratory – LNLS/Sirius Status Report

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Computing Platforms (COMP)

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MXCuBE status

- Versions: Backend | Database & Frontend
- Production Version:
Web MXCuBE3 - Backend and Frontend
- Development Version:
Web MXCuBE3, Web MXCuBE4, mxcubecore

Developments since last meeting

- Improving: LNLSCollect Class
 - Creation and updating of the snapshot system before collections begin.
 - Improvements in logs, error handling, and attempts to prevent scans from crashing.
 - Debugging the acquisition loop.

- BlueSky with MXCuBE in LNLS:
 - Update of the LNLSCollect class for using bluesky plans.
 - Integration of the Http-server API for running plans.
 - Integration with a Kafka instance for the live monitoring of the plans.

- Updated and cleaned up:
 - LNLSDetDistMotor Class
 - Update moves with granite.
 - LNLSCollect Class
 - Snapshot proceed before scan.
 - LNLSMachInfo Class
 - Update of XML files to reflect new machine configurations.
- Tests, CI/CD and Code Quality:
 - Integration with SonarQube and more jobs.

Parallel Developments

- An automatic pipeline for multiple crystals experiments has been implemented. It uses XDS, CCP4 and Phenix.
- An automatic pipeline for synchrotron serial crystallography (SSX) is under development. It uses CrystFEL.
- A web application for data processing has been developed and deployed.

Parallel Developments – Data Processing

- MANACAutoproc Web: web application where users can access and process their data.
- This app also allows the users to visualize their diffraction data using H5web components, as well as model their processed data with Moorhen.

MANACAutoproc Web

Display Documentation

Login

Username



Password



Macromolecules



Small Molecules

Login

Parallel Developments – Image Viewer

Search Files

/ibira/lnls/beamlines/manaca/proposals/20210016/data/20230726/teste-1-09/lys_0001_00001.cbf

Choose Image:

1

Previous

Next

Current Image: 1

Number of Images: 1800

Fast-Forward Stride Value:

10

Stop

Fast-Forward

Add Resolution:

☒ Show Resolution Rings

4.41 Å

Remove

2.32 Å

Remove

1.67 Å

Remove

1.36 Å

Remove

1.19 Å

Remove

Set Threshold:

32

☒ Linear

☐ Log

☐ SymLog

☐ Sqrt

☒ Invert

Map Color:

☒ Greys

☐ Inferno

☐ Rainbow

☐ Viridis

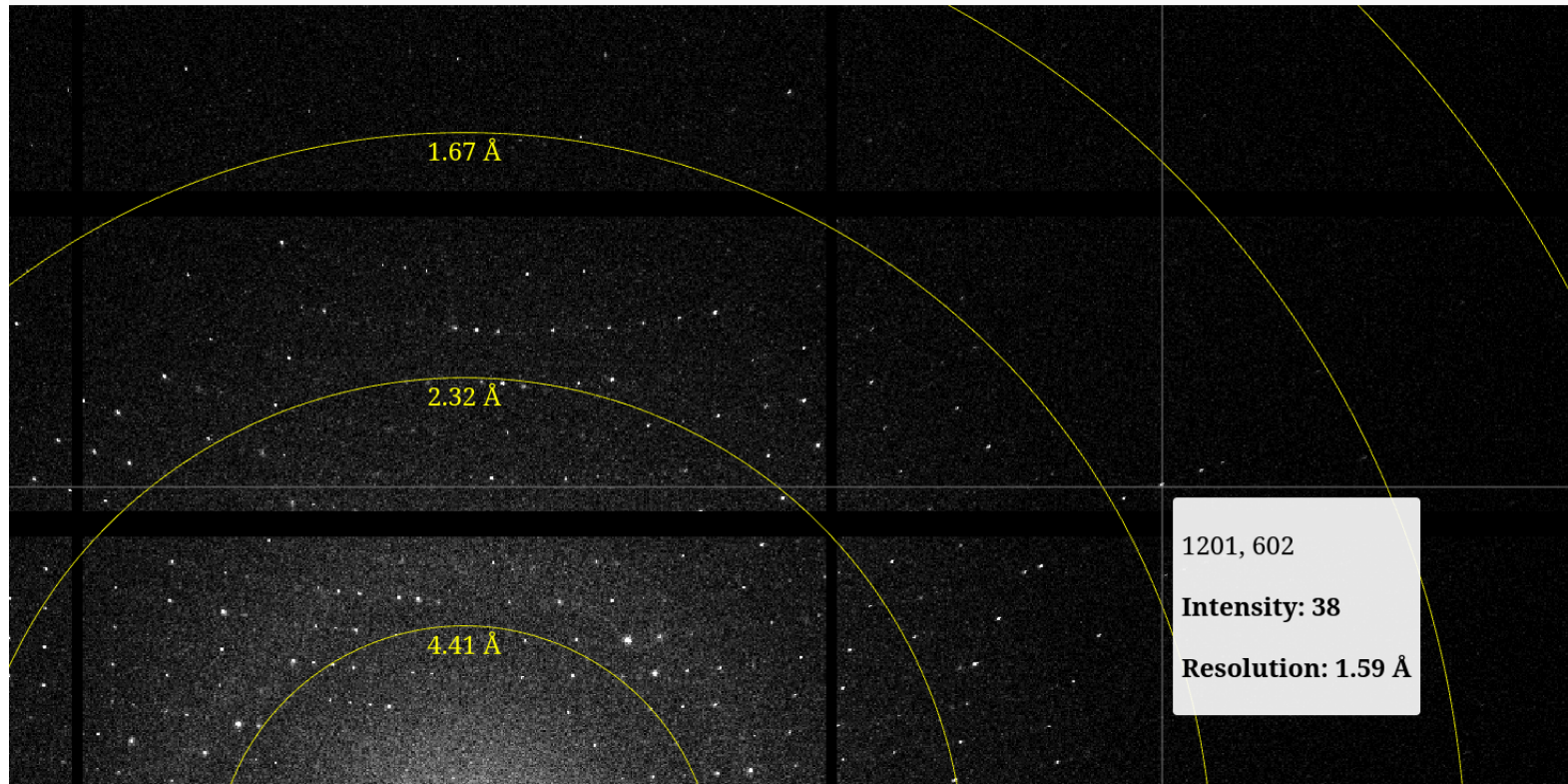
Peak Search

Find Spots

a: 0 b: 0 c: 0

alpha: 0 beta: 0 gamma: 0

☒ Show Peaks



1.67 Å

2.32 Å

4.41 Å

1201, 602

Intensity: 38

Resolution: 1.59 Å

30

25

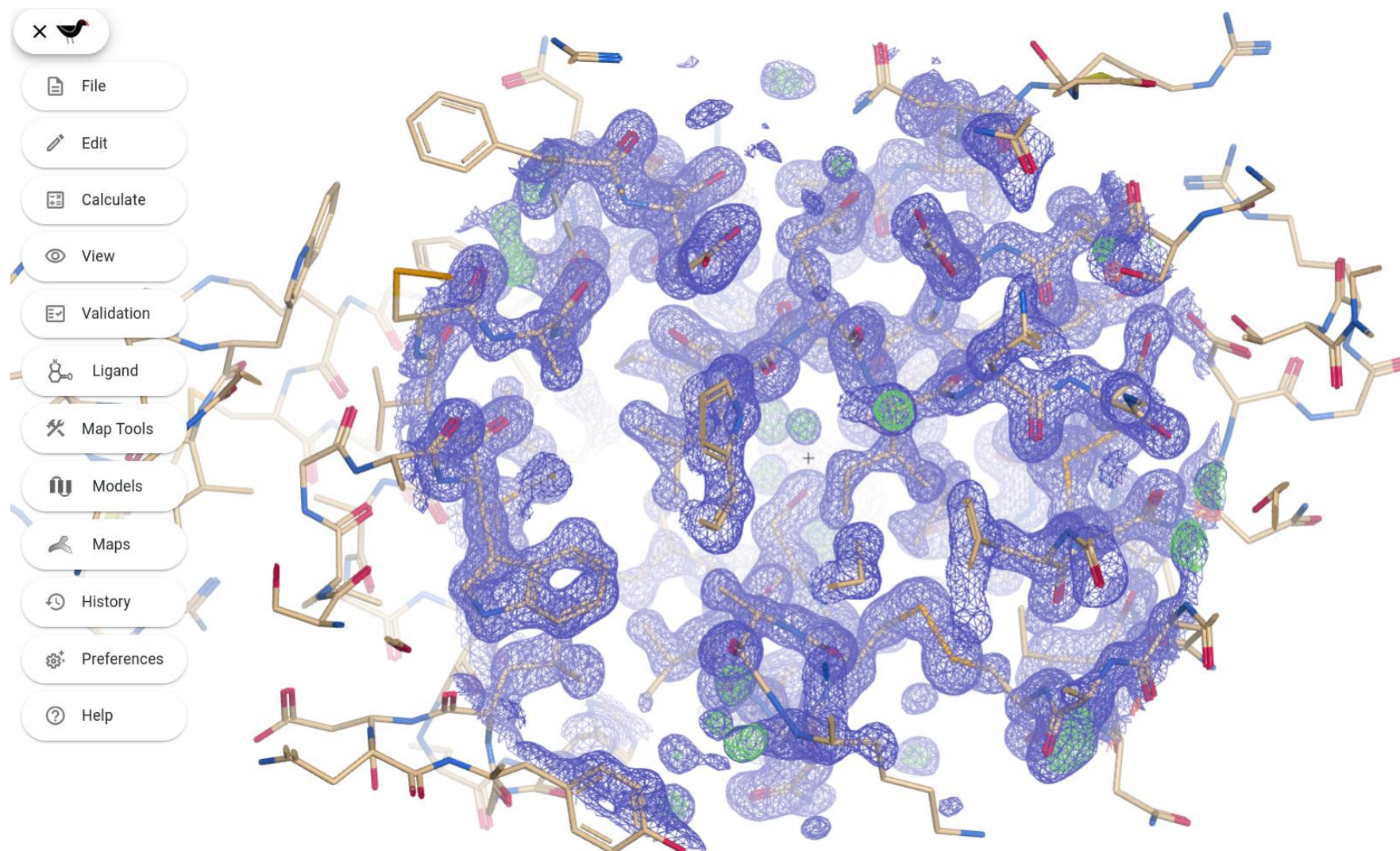
20

15

Parallel Developments – Image Viewer

- Used for visualizing data collected at the beamline.
- Uses H5Web.
- Inspired by Braggy from ESRF.

Parallel Developments – Moorhen



Parallel Developments – Moorhen

- Moorhen is a web version of coot, created by a team of developers in the UK.
- A local instance of Moorhen has been implemented and connected to our database.
- Users can easily access and model the data that has been processed at the beamline.

Plans for the next six months

- Update of LNLSCollect and complete integration with Bluesky.
- Integrate new equipments and scans, refactor LNLS files.
- Test MXCuBE 4 with new scan routines.

Thank you

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